

Detector Support Context

Component-guided preprocessing isolates wire geometry before any connectivity reasoning begins.

Input scan

Scanned or
hand-drawn
schematic



Component priors

Filename-matched
HDC labels and
oriented boxes



Occlude + crop

Polygon fill and
component-union
ROI selection



Sauvola cleanup

Adaptive thresholding
plus light
morphological closing



Wire fragments

Connected traces,
PCA endpoints,
and anchor filtering

Main Contribution: Join Recovery

Endpoint graph + guarded completion + netlist verification

Endpoint graph

Pin, endpoint,
and T-junction
hypotheses



Degree-budget completion

Floating-pin recovery
through min-cost
b-matching



Netlist output

Recovered component
connectivity and
SPICE structure

Measured detector quality feeds the join stage; the join stage determines whether those fragments become credible electrical connectivity.